EC Sensor (V2.1) — User Guide

How it works, pin configuration, calibration, and target EC ranges

1. How EC Sensing Works

EC stands for **Electrical Conductivity** — a measure of how well a liquid conducts electricity. In water, conductivity comes from dissolved salts and nutrients (called ions). More dissolved nutrients means more ions, which means the water conducts electricity more easily — so a higher EC reading.

Your sensor works by passing a small AC (alternating current) signal through two electrodes on the probe tip, submerged in the liquid. The circuit on the board (built around an op-amp chip and an analog switch chip) measures how easily that signal passes through the liquid between the electrodes, and converts that into a voltage. An AC signal (instead of DC) is used specifically to prevent the electrodes from corroding over time — this is called avoiding "polarization."

That voltage is what your microcontroller (ESP32) reads on the analog pin. On its own, this voltage is just a number — it only becomes a meaningful EC value once the sensor has been **calibrated** (see Section 4).

2. Pin Configuration

Your board is labeled **"EC V2.1"** — a design based on the same principle as DFRobot's official Gravity Analog EC Sensor (DFR0300), with one added pin. It has 4 header pins:

Pin	Function	Connect to
A0	Analog EC signal output (0–3.4V)	ESP32 GPIO34 (ADC1, input-only, WiFi-safe)
T1	External temperature probe data line (pass-through, unused unless a DS18B20 temperature probe is added)	Leave unconnected for now
GND	Ground	ESP32 GND
VCC	Power input (3.0–5.0V)	ESP32 3.3V

Note: Only A0, GND, and VCC need to be wired for EC readings to work. T1 can stay disconnected — it will not cause errors. Without it, temperature compensation is skipped and a fixed value (25°C) is assumed in code instead.

3. Basic Usage

1. Wire the sensor: A0 → ESP32 GPIO34, VCC → 3.3V, GND → GND. Leave T1 unconnected.
2. Screw the EC probe into the BNC connector on the board if not already attached.
3. Upload firmware using the DFRobot_EC library (or an equivalent EC-conversion function) to read the analog voltage on A0 and convert it to an EC value in mS/cm.

4. Before trusting any reading, calibrate the sensor once using standard buffer solutions — see Section 4.
5. Rinse the probe with distilled water before and after each use in a different liquid, to avoid cross-contaminating your readings.

4. Calibration Steps

Calibration teaches the sensor what voltage corresponds to a known, exact EC value — like setting a bathroom scale to zero before weighing something. Two calibration points let the sensor draw an accurate line between them, so it can correctly convert any future voltage reading into a real EC value.

Step 1: Open Serial Monitor

Upload the sketch, then open the Arduino IDE Serial Monitor and set the baud rate to 115200 (must match the code).

Step 2: Enter calibration mode

Type **enterec** and press Enter. This pauses normal measuring and switches into calibration mode.

Step 3: Dip in the first solution

Rinse the probe with distilled water, dry it, then fully submerge it in the **1413 µS/cm** buffer solution. Wait 10–20 seconds for the reading to settle.

Step 4: Calibrate point 1

With the probe still in the solution, type **calec** and press Enter. The library recognizes the voltage range and calibrates this point automatically.

Step 5: Dip in the second solution

Rinse and dry the probe again, then submerge it in the **12.88 mS/cm** buffer solution. Wait for the reading to settle.

Step 6: Calibrate point 2

Type **calec** again. The library detects this is the higher-range solution and calibrates the second point — no need to specify which solution it is.

Step 7: Save and exit

Type **exitec** and press Enter. This permanently saves the calibration values and returns to normal measuring mode.

Tip: if you don't have buffer solutions yet, you can skip calibration temporarily and still use the sensor for relative comparisons (e.g. "is EC rising or falling") — just don't treat the raw numbers as lab-accurate until calibrated.

5. What Do "1413 µS/cm" and "12.88 mS/cm" Mean?

These are the exact conductivity values of two **standard buffer solutions** — liquids manufactured to a precise, known conductivity, sold specifically for calibrating EC sensors.

µS/cm (microsiemens per centimeter) and **mS/cm** (millisiemens per centimeter) are both units of conductivity — mS/cm is simply 1,000× larger than µS/cm. So 12.88 mS/cm is the same as 12,880 µS/cm.

Value	What it represents
1413 µS/cm	A low-conductivity reference point — roughly similar to moderately mineralized drinking water.
12.88 mS/cm (= 12,880 µS/cm)	A high-conductivity reference point — roughly similar to a concentrated nutrient solution or brackish water.

By dipping the probe into each one and telling the sensor "this is exactly 1413" and "this is exactly 12,880," it learns the correct voltage-to-EC relationship for your specific probe.

6. Good EC Value Ranges (Hydroponics Reference)

EC tells you how nutrient-concentrated a solution is. Too low and plants go hungry; too high and roots get stressed or burned. These are general starting ranges — always check the specific needs of your crop.

Stage / Plant Type	EC Range (mS/cm)
Seedlings / young plants	0.5 – 1.0
Leafy greens (lettuce, herbs)	1.0 – 1.6
Fruiting plants (tomato, cucumber, pepper)	1.8 – 3.0
Strawberries	1.2 – 1.8
General-purpose hydroponics (safe default)	1.2 – 2.5

Quick rules of thumb:

- Below ~0.8 mS/cm → solution too weak, plants likely nutrient-deficient.
- Above ~3.5 mS/cm → solution too strong, risk of root burn or salt stress.
- Start low and increase gradually rather than over-dosing nutrients early.
- In recirculating systems, EC tends to rise over time as plants absorb water faster than nutrients — dilute with plain water if it climbs too high.
- Always compare readings taken at similar temperatures, since EC naturally rises with solution temperature.